Fiscal Year:	FY 2013	**Task Last Updated:**	FY 07/02/2013
PI Name:	Durian, Douglas J. Ph.D.		
Project Title:	The Melting of Aqueous Foams		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6396	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	1995 Microgravity Fluid Physics NRA-94-OLMSA-05: Research and Flight Experiment Opportunities
Start Date:	08/05/2004	**End Date:**	06/30/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
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Flight Program:			
Flight Assignment:	NOTE: End date now 6/30/2013, per NSSC (Ed., 6/17/13) NOTE: Grant number was changed in 2007 to NNX07AP20G (Ed., 1/4/13) NOTE: Funding extended to 4/04/2013 per NSSC database (5/2010) NOTE: Funding extended to 4/4/2010 through a new grant in 2007 (6/2008) NOTE: Funding reinstated thru 9/30/2008, per F. Chiaramonte (8/2006). Suppl funding thru Oct 2006 (6-5-06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX07AP20G ; NNX07AK60G ; NNC04GB61G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

The unusual elastic character of foams will be quantified macroscopically by measurement of the shear stress as a function of static shear strain, shear strain rate, and time following a step strain; such data will be analyzed in terms of a yield stress, a static shear modulus, and dynamical time scales. Microscopic information about bubble packing and rearrangement dynamics, from which these macroscopic non-Newtonian properties presumably arise, will be obtained non-invasively by novel multiple-light scattering diagnostics such as diffusing-wave spectroscopy (DWS) and speckle-visibility spectroscopy (SVS). Bubble size and homogeneity will be studied by diffusely transmitted light (DTS). Liquid fraction will be measured by electrical conductivity. Quantitative trends with materials parameters, such as average bubble size, and liquid content, will be sought in order to elucidate the fundamental connection between the microscopic structure and dynamics and the macroscopic rheology.

There are three all-encompassing objectives motivating the FOAM project. The first concerns multiphase flows and represents the “mechanics” aspect alluded to in the F.O.A.M. acronym: To understand how foams and bubbly liquids deform and flow in response to applied forces. The second objective concerns multiphase separations: To understand how foams and bubbly liquids evolve with time via a combination of liquid flow between bubbles, of gas diffusion across the film separating neighboring bubbles, and of direct coalescence via film rupture. For both flows and separations, some level of understanding has been established for very dry foams, and also for liquids with relatively few bubbles; however, we are more ignorant of the crucial intermediate regime of very wet foams and very bubbly liquids. This regime defies earth-bound experiments. The third, overall, objective is to transfer this new understanding of multiphase flows and separations into ASL technology for NASA's strategic exploration mission.

The utility and fascination of foams are derived largely from the surprising fact that they have a solid-like elastic character in spite of being mostly gas with a few percent volume fraction of liquid, but can nevertheless flow under shear. The physical origin of such unusual rheology in terms of microscopic structure and dynamics is poorly understood and remains a subject of basic scientific interest to physicists, chemists, and chemical engineers. The proposed research promises important new insight into these issues, and could also have significant consequences for our understanding of flow in other dense randomly-packed systems such as emulsions, colloidal suspensions, slurries, bubbly liquids, and granular materials. Furthermore, all foam applications are empirically based and the proposed research may generate valuable fundamental guidance for the development of materials with more desirable rheology and better stability.

The value of the FOAM project in terms of NASA’s exploration mission lies with Advanced Life Support. Water recovery and recycling are absolutely central to ASL, and require fine control of multiphase flows and separations. The very heart of the FOAM project is to bring new technology and understanding to these problems.

NOTE: Continued by "FOAM--Foam Optics and Mechanics (The Melting of Aqueous Foams)," grant NNX14AM99G with the same Principal Investigator, Douglas Durian, Ph.D.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Foams are important for a clean environment. Firstly, their presence in natural waters signals the presence of surface-active impurities and a need for clean-up. Secondly, they are useful themselves in the cleanup process since many impurities can be removed by injecting gas and skimming off the resulting foam. Relatedly, study of foams brings specific insight into multiphase flow and separation phenomena of direct importance for water recovery and recycling operations.

Task Progress:

SUPPLEMENTAL INFORMATION FOR FINAL REPORTING (JULY 2013)
Since the previous task report in January 2013, my group consisted of senior graduate student Adam Roth (funded by the grant) and first-year graduate student Anthony Chieco (unfunded). Together we made progress on the following:

[1] Finalize a long publication on the structure, statistics, and coarsening rate of wet quasi-2d foams. This paper introduces a novel apparatus for controlling the wetness of the foams, includes extensive data on bubble sizes, side-number statistics, shapes, and size-topology correlations. It also includes extensive data for coarsening rates, and comparison with a model that we construct based on the total blockage of gas flux by the Plateau borders. This paper is now in print.

[2] Prepare and submit an article on similar measures for the "surface foam" consisting of the faces of bubbles in a 3d bulk sample pressed against the boundary of the sample. This paper introduces a novel optical method for isolating the surface faces, totally eliminating the bulk structure of the foam. This paper is now submitted and is under review.

[3] Analyze existing data for several measures of hyper uniformity, in collaboration with Sal Torquato, and write up a complete draft. This paper should be ready for submission soon.

[4] Explore a new method for analyzing image data, based on a Monte Carlo technique. Currently this is being done for 2d foams, with the idea being to generalize to 3d tomography data. The great advantage is that it allows the known microstructure of Plateau border junctions to be built-in and hence should greatly aid analysis of larger and less-perfect images.

REPORTING FOR ANNUAL PROGRESS FROM APRIL 2012-JANUARY 2013:

During the current period, my group consisted of senior graduate student Adam Roth (funded by the grant) and first-year graduate student Anthony Chieco (not funded). Together we made the following progress:

[1] Prepare publication on the coarsening of foam on a curved substrate, which alters vonNeumann's law according to a theory that we tested. Shepherd the manuscript through a round of peer-review, and ultimate acceptance for publication.

[2] Construct a border-blocking model for the coarsening of 2d wet foams, which depends on two particular measures of bubble shape that we introduce. Analyze image data for bubble sizes, shapes, and topology, and compare with our prediction.

	[3] Acquire high-quality data for the surface structure of 3d foams, using a custom apparatus that takes advantage of total-internal reflection effects to eliminate contamination of the image by bulk structure. Begin to analyze the data, as above for the 2d foam. [4] Continue work on reconstruction of foam structure from 3d grayscale tomography data previously acquired using a custom apparatus. [5] Begin a project to analyze the 3d grayscale data for the structure factor, as a means to test for hyper uniformity.
Bibliography Type:	Description: (Last Updated: 06/12/2025)
Articles in Other Journals or Periodicals	Roth AE, Jones CD, Durian DJ. "Bubble statistics and coarsening dynamics for quasi-two dimensional foams with increasing liquid content." arXiv.org: e-prints in Physics, Mathematics, Computer Science, Quantitative Biology, Quantitative Finance and Statistics. June 2012. http://arxiv.org/abs/1207.0146 , Jun-2012
Articles in Other Journals or Periodicals	Roth AE, Chen BG, Durian DJ. "Structure and coarsening at the surface of a dry three-dimensional aqueous foam." arXiv.org: e-prints in Physics, Mathematics, Computer Science, Quantitative Biology, Quantitative Finance and Statistics. June 2013. http://arxiv.org/abs/1306.2939 (preprint). , Jun-2013
Articles in Peer-reviewed Journals	Roth AE, Chen BG, Durian DJ. "Structure and coarsening at the surface of a dry three-dimensional aqueous foam." Physical Review E. 2013 Dec;88(6):062302. https://doi.org/10.1103/PhysRevE.88.062302 , Dec-2013
Articles in Peer-reviewed Journals	Roth AE, Jones CD, Durian DJ. "Coarsening of a two-dimensional foam on a dome." Physical Review E. 2012 Aug;86(2 Pt 1):021402. Epub 2012 Aug 6. http://dx.doi.org/10.1103/PhysRevE.86.021402 ; PubMed PMID: 23005758 , Aug-2012
Articles in Peer-reviewed Journals	Roth AE, Jones CD, Durian DJ. "Bubble statistics and coarsening dynamics for quasi-two dimensional foams with increasing liquid content." Physical Review E. 2013 Apr;87(4):042304. (14 p.) http://dx.doi.org/10.1103/PhysRevE.87.042304 , Apr-2013
Dissertations and Theses	Roth AE. "Structure and Coarsening of Foams: Beyond von Neumann's Law." Dissertation, University of Pennsylvania, May 2013. , May-2013
Significant Media Coverage	Roth AE, Jones CD, Durian DJ. "Selection of Figure 1 of article 'Coarsening of a two-dimensional foam on a dome,' Physical Review E 86, 021402 (2012), by editors for on-line 'Kaleidoscope of Images.' " Kaleidoscope of Images, Physical Review E, http://pre.aps.org/kaleidoscope/pre/86/2/021402 , Aug-2012